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Cognitive-Behavior Therapy for Disaster-Exposed Youth With Posttraumatic Stress: Results From a Multiple-Baseline Examination

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Youth traumatized by natural disasters report high levels of posttraumatic stress such as symptoms of posttraumatic stress disorder, other anxiety disorders, and depression. Research suggests that cognitive behavioral therapies are promising interventions for symptom reduction; however, few cognitive behavioral treatments have been systematically tested in youth hurricane survivors. The current study provides an examination of the efficacy of an intervention manual designed specifically for hurricane-exposed youth (i.e., the *StArT* manual) using a partially nonconcurrent multiple baseline design. Youth ages 8–13 (n=6) who met diagnostic criteria for posttraumatic stress disorder were provided the individual *StArT* treatment in their school. Youth were assessed at pretreatment, weekly during treatment, and at posttreatment. Results provide initial evidence for the efficacy of the *StArT* manual and suggest the feasibility of conducting the *StArT* manual in a school setting. The importance of large-scale tests of effectiveness and implementation of cognitive behavioral treatments in the wake of disaster among youth are discussed.

A SUBSTANTIAL PORTION OF youth are exposed to traumatic events (Costello, Erkanalli, Fairbank, & Angold, 2002) such as abuse (Ackerman, Newton, McPherson, Jones, & Dykman, 1998), violence (Stein, Jaycox, Kataoka, Rhodes, & Vestal, 2003), and disasters (La Greca, Silverman, Vernberg, & Roberts, 2002; Weems & Overstreet, 2008). This is concerning given that youth exposed to traumatic stress can suffer from an array of impairing emotional and behavioral problems. Youth responses to trauma include the development of posttraumatic stress disorder (PTSD), which is characterized by high levels of negative reexperiencing, hyperarousal, emotional numbing, and avoidance (DSM-IV–TR; American Psychiatric Association, 2000), other anxiety disorders, and depression. In terms of long-lasting effects, the experience of childhood trauma increases the vulnerability for developing substance abuse problems (Copeland, Keeler, Angold, & Costello, 2007), chronic PTSD (Widom, 1999), other anxiety disorders (e.g., generalized anxiety disorder [GAD], social anxiety disorder; Copeland et al., 2007), depression (Bolton et al., 2004; Copeland et al., 2007), and disruptive behavior disorders (Copeland et al., 2007). Further, adults with childhood trauma histories evidence decreased likelihood for obtaining successful employment or graduating from high school (McGloin & Widom, 2001).

Given the potential lasting impact of trauma (Weems et al., 2010), effective interventions are paramount for promoting positive adjustment in traumatized youth. Cognitive behavioral therapies (CBTs) have garnered positive empirical support (e.g., Chemtob, Nakashima, & Hamada, 2002;

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King et al., 2000; Stein, Jaycox, Kataoka, Wong, et al., 2003; see Silverman et al., 2008, for review and meta-analysis), are typically exposure based, and include various additional specific techniques such as psychoeducation, cognitive coping strategies, and relapse prevention (see Silverman et al., 2008). CBTs have been shown to result in the reduction of posttraumatic stress levels in youth exposed to several different types of traumas (e.g., disaster: Chemtob et al., 2002; abuse: King et al., 2000; community violence: Stein, Jaycox, Kataoka, Wong, et al., 2003) and are considered to be an efficacious treatment for traumatized youth (Silverman et al., 2008).

Despite promising evidence, the odds are against many traumatized youth in terms of receiving an effective intervention. The mental health needs of many children and adolescents, particularly those who are uninsured or from low-income families, often go untreated (Kataoka, Zhang, & Wells, 2002). Youth may also encounter additional barriers to receiving treatment in traditional settings, such as lack of transportation, stigma, or child care (Stephan, Weist, Kataoka, Adelsheim, & Mills, 2007). School-based implementation of CBT can provide youth greater accessibility to treatment than traditional settings (Evans, 1999), have been shown to reduce stigma associated with seeking help (Nabors, Weist, & Reynolds, 2000), offer enhanced opportunities for generalization and maintenance of treatment goals (Evans, 1999), and allow interventionists to reach large numbers of at-risk youth (Levitt, Saka, Hunter-Romanelli, & Hoagwood, 2007). Given these potentials, there has been growing interest in offering CBT in school, particularly for youth exposed to natural disaster-related traumatic stress (e.g., Chemtob et al., 2002; Salloum & Overstreet, 2008).

Research suggests that youth exposed to hurricane-related traumatic stress can develop high levels of emotional distress (e.g., La Greca, Silverman, Vernberg, & Prinstein, 1996; Lonigan, Shannon, Taylor, Finch, & Sallee, 1994). The research to date for disaster-exposed youth indicate reductions in youths' posttraumatic stress symptoms from pre- to posttreatment, as well as maintenance of treatment gains at follow-up (Chemtob et al., 2002; Salloum & Overstreet, 2008). To date, however, relatively few studies have investigated the efficacy of school-based treatments for youth exposed to natural disasters (compared to other types of traumas) and most have taken place in the relatively recent aftermath of the disaster (e.g., earthquakes: Goenjian et al., 2005; hurricanes: Chemtob et al., 2002; Salloum & Overstreet, 2008; tsunami: Catani et al., 2009). Indeed, few treatment manuals exist and no treatments for hurricane-exposed youth have undergone an idiographic test of efficacy (see Silverman et al., 2008; Taylor & Chemtob, 2004).

The *StArT* manual (Saltzman et al., 2007) is a treatment manual for hurricane-exposed youth that was developed by a group sponsored by the National Child Traumatic Stress Network (NCTSN) of the National Center for PTSD (2006; Allen, Saltzman, Brymer, Oshri, & Silverman, 2006). *StArT* is a hurricane trauma-focused CBT that has multiple empirically supported components or modules. While the components are empirically based, no studies have specifically tested the *StArT* manual (Allen et al., 2006). Efficacy evaluation of the *StArT* manual (Allen et al., 2006; Saltzman et al., 2007) has the potential to make an important contribution to the disasters intervention database. Youth exposed to Hurricane Katrina-related traumatic stress, for example, have been shown to be at risk for long-term emotional problems (Weems et al., 2010). Hurricane-affected areas often remain in clear need of mental health services for many years following the disaster (Drury, Scheeringa, & Zeanah, 2008; Weems, 2010), and in particular youth exposed to Hurricane Katrina, perhaps because contextual factors (e.g., damage and disrepair) contribute to PTSD symptom stability (Weems et al., 2010). Thus, initial support for the *StArT* manual may foster wider application and empirical tests of the intervention following future disasters.

In this study, the goal was also to answer the call for the increased use of idiographic strategies in intervention research (Barlow & Nock, 2009) to provide initial support for the *StArT* manual. While extant research with disaster-exposed youth in school settings shows promise, these intervention studies have investigated symptom reduction through nomothetic designs (e.g., Chemtob et al., 2002; Salloum & Overstreet, 2008). A limitation of group comparison studies is the difficulty in identifying individual variation in treatment-related symptom change and individual variation in response to treatment components (Barlow & Nock, 2009; Kazdin, 2008). The use of multiple baseline (MBL) designs, for example, can help identify the pattern of symptom reduction across treatment sessions to isolate individual differences in treatment response (Kazdin, 2003) and have provided critical evidence of intervention efficacy in the child anxiety disorders (e.g., Lumpkin, Silverman, Weems, Markham, & Kurtines, 2002; Ollendick, 1995) and PTSD treatment literature (e.g., Feather & Ronan, 2006; Saigh, 1987).

In summary, the aim of this study was to provide an initial evaluation of the efficacy of the *StArT* manual for reducing PTSD symptoms. A partially

Table 1
Pre- and Posttreatment ADIS Data

Child	Pretreatment		Posttreatment	
	Diagnosis	CSR	Diagnosis	CSR
Jennifer	OCD[a]	7	None	N/A
	SP[b]	7		
	GAD[c]	6		
	PTSD[d]	4		
	SAD[e]	4		
John	PTSD[d]	8	None	N/A
	GAD[c]	8		
	Dysthymia	8		
	OCD[a]	6		
	ADHD[f]	4		
Michael	SAD[e]	7	SP[b]	7
	SP[b]	7	Social[g]	4
	GAD[c]	7		
	PTSD[d]	4		
	ADHD[f]	4		
Sarah	PTSD[d]	7	SP[b]	7
	GAD[c]	7	Dysthymia	5
	Dysthymia	7	GAD[c]	4
	ADHD[f]	7		
Elizabeth	SR[h]	8	GAD[c]	8
	ADHD[f]	8	ADHD[f]	8
	GAD[c]	7	Social[g]	4
	Social[g]	7		
	SAD[e]	5		
	PTSD[d]	4		
Kelly	Dysthymia	8	None	N/A
	SAD[e]	6		
	SP[b]	6		
	GAD[c]	6		
	PTSD[d]	4		

Note. [a]Obsessive compulsive disorder; [b]Specific phobia; [c]Generalized anxiety disorder; [d]Posttraumatic stress disorder; [e]Separation anxiety disorder; [f]Attention-hyperactivity deficit disorder; [g]Social phobia; [h]School refusal diagnosis.

nonconcurrent MBL was used to identify individual- and session- (and/or manual module) level change in PTSD symptoms. PTSD symptoms were expected to show relative stability over baseline, with reductions at the implementation of therapy and over the course of treatment in a manner similar to previous MBL designs with PTSD (e.g., Feather & Ronan, 2006; Saigh, 1987) in particular, and similar to previous MBL designs with childhood anxiety disorders (e.g., Lumpkin et al., 2002; Ollendick, 1995) in general. In addition to examining individual PTSD symptom trajectories over the course of treatment, individual- and group-level changes in related symptoms and problems (e.g., cognitive errors, anxiety control, anxious and depressive symptoms) were examined using multitrait (anxiety symptoms, cognitive errors) and multi-informant (child and parent clinical interviews) data collection. It was hypothesized that the intervention would improve symptom levels and reduce diagnoses/impairment.

Method

PARTICIPANTS

All participants were exposed to Hurricane Katrina and/or its aftermath and were students at public schools in New Orleans. These schools are located in neighborhoods that received massive hurricane damage, almost total flooding, and continue to have a significant amount of disrepair. Youth attending are predominantly from low-income families (school data indicate that 97% of the students receive free lunch, 2% are on a reduced payment, and 1% pay for lunch). Through collaboration with these schools, the University of New Orleans Youth and Family Anxiety, Stress, and Phobia Laboratory provided school-based screenings to students as an adjunct to traditional school counseling. Youth recruited for this intervention project reported total scores on the Posttraumatic Stress Disorder Reaction Index (RI) in the severe to very severe range (see Measures section for a description of RI cutoff scores) at a schoolwide screening assessment and the school counselors provided parent contact information ($n = 21$). Of these, 15 families were able to be reached by the contact information and 13 consented to a diagnostic evaluation. Seven of these 13 youth met inclusion criteria for this study (i.e., met PTSD diagnosis upon diagnostic evaluation, see below). The *StArT* intervention was implemented with four youth in spring 2009 and three youth in fall 2009. One child left treatment after the first session (fall 2009) and was not included in this study. The family decided they did not want to participate and referral information was provided to the family. To maintain anonymity and for convenience of description, common first names were assigned to each participant. A brief description of the child exposure experiences and presenting problems are provided below. Child report of pretreatment clinical diagnoses and their severity are presented in Table 1; pretreatment posttraumatic stress symptom levels are shown in Figure 1.

Detailed Child and Family Characteristics

Jennifer. Jennifer is a 13-year-old African American girl who resides with her mother, stepfather, and four younger siblings. She evacuated New Orleans with her mother and extended family, leaving her pets behind. The family reports a prolonged evacuation experience that included encountering several hours of traffic upon leaving the city. Once 200 miles outside the city, the family sought accommodations at restaurants and hotels, and reported that they were denied services due to discrimination. In addition to these stressful evacuation experiences, Jennifer reported missing possessions lost due to the hurricane, and her cats that were left behind. The family also moved and so she lost contact with friends from her old school. She received treatment during spring 2009.

John. John is a 9-year-old African American boy. He and his younger sister reside with his cousin, who is their legal guardian. John and his sister were removed from their mother's custody due to neglect. He came into his cousin's custody the year of Hurricane Katrina. He, his sister, and his cousin did not evacuate for the storm. According to John, he witnessed rising floodwaters and saw someone drown during the storm. The family's home flooded, but they managed to escape to one of the interstate bridges and were rescued by helicopter. They have since rebuilt their home. At the initial interview, John reported having bad dreams about the hurricane, frequently worrying about his cousin's welfare, or about his own personal safety. He received treatment during spring 2009.

Michael. Michael is an 8-year-old African American boy residing with his mother and older brother. Though the family was able to evacuate for the hurricane, their home was severely damaged and their possessions were destroyed. He reports disturbing memories of hurricane-related television coverage and from seeing the city for the first time after the storm. When his family returned to assess home damages after the storm, he saw the decomposed remains of his grandmother's dog. At the initial interview, his mother indicated that Michael was fearful of and worrisome during separation from her, and that these symptoms have worsened since her recent divorce from Michael's father. He received treatment during spring 2009.

Sarah. Sarah is a 12-year-old African American girl who resides with her older sister and her legal guardians (her paternal aunt and uncle; her mother passed away when Sarah was 5, and she and her siblings have been in their custody since then). She evacuated with her sister and her aunt prior to the hurricane. They left behind most of their possessions and pets. Her uncle left a day later than the rest of the family. He could not be reached for 4 days after the storm and family members were unsure of his whereabouts. While Sarah's guardian reports that she and her siblings handled the evacuation "well," Sarah reports being very upset by news coverage of the storm and reports continued grief over lost possessions. The family home took in 4 feet of water and rebuilding began upon their return to the city. She received treatment during fall 2009.

Elizabeth. Elizabeth is a 9-year-old African American girl who resides with her mother and older half sister. Before the hurricane made landfall, she and her family evacuated to the northern part of the state and remained there for a month before moving back to the greater New Orleans area. Within 2 months of the storm, Elizabeth's father passed away. According to her mother, Elizabeth frequently complained of heart attack-like symptoms and other types of somatic symptoms (i.e., stomachaches and headaches), and repeatedly visited the emergency room for these symptoms during this time. She continued to report somatic complaints as well as difficulty sleeping at the initial interview. She received treatment during fall 2009.

Kelly. Kelly is an 8-year-old African American girl who resides with her mother. Their family evacuated New Orleans with a car full of their belongings before hurricane landfall. The remaining possessions, including the child's fish, were destroyed during the storm. Kelly experienced prolonged evacuation from the city. She and her relatives did not return to the city until 9 months after the storm. During the evacuation period, she reported being lonely, not having many friends at her new school, and missing friends from her old school. At the initial interview, she became tearful when talking about Katrina, and reported worrying that her grandmother would lose her home again when Hurricane Gustav came in 2008. During the initial interview, Kelly indicated worries relative to her family's safety, and stated that the sight of homes still in disrepair made it difficult for her to stop thinking about Hurricane Katrina. She received treatment during spring 2009.

MEASURES

The Anxiety Disorders Interview Schedule for DSM-IV—child and parent versions (ADIS-IV; Silverman & Albano, 1996) was administered in full to children and in part to their parents to determine diagnosis and as part of posttreatment assessment. The ADIS-IV is a semistructured interview designed for youth ages 7 to 17. The ADIS-IV provides assessment of childhood anxiety disorders, affective disorders (i.e., dysthmia, major depressive disorder), and externalizing disorders (i.e., attention-deficit hyperactivity disorder [ADHD], conduct disorder, oppositional defiant disorder), as well as sections for the assessment of other symptoms and impairment commonly experienced in childhood (i.e., school refusal behavior, psychosis, selective mutism, eating disorders, and somatoform disorders). Items on the ADIS PTSD scale include inquiry about specific types of potentially traumatic events experienced (e.g., car accidents, fires, abuse, natural disasters) followed by a series of DSM-based questions regarding the types of PTSD symptoms experiences relative to trauma. For this study, youth provided self-reported trauma history, and were queried about posttraumatic stress symptoms related to Hurricane Katrina.

The parent and child versions of the ADIS-IV have been evaluated for test–retest reliability. For the child version, intraclass coefficients (ICCs) were reported in the excellent range (ICC = 0.85–0.92) for children ages 7 to 11 and for youth ages 12 to 16 (ICC = 0.81–0.99; Silverman, Saavedra, & Pina, 2001). For the parent version, reliability estimates have been reported in the excellent range for the younger group (ICC = 0.86–0.99), and in the good-to-excellent range for the older group (ICC = 0.52–0.94; Silverman et al., 2001). Parent and child interview schedules are sensitive to treatment effects (Kendall, 1994; Silverman, Kurtines, Ginsburg, Weems, Lumpkin, et al., 1999). At pretreatment, the therapist (the first author) conducted the ADIS-IV with children at their schools.

Because a goal was to increase participation and conducting an ADIS-IV with parents would be difficult in a school setting (e.g., scheduling; administration of the parent version can take a couple of hours) only the PTSD scale was administered to parents and was conducted over the phone. ADIS-IV PTSD scale information obtained from children and their parents was used to assign a clinical severity rating (CSR) for PTSD diagnoses. CSRs range from 0 to 8, and ratings of 4 or greater are indicative of clinical diagnosis. Thus, youth and parents reporting CSRs of 4 or greater on the PTSD scale were included in this study. Because parents were not administered the entire ADIS, the parent report could not be used when assigning CSRs for diagnoses other than PTSD. CSRs were assigned to non-PTSD diagnoses based on child report only. To limit the influence of demand characteristics on responses to interview items, children's pre- and posttreatment interviews were not conducted by the same interviewer and the postinterview was not conducted by the therapist. Though interviewers were aware that youth had been exposed to Hurricane Katrina, they were blind to children's pretreatment diagnoses. ADIS-IV interviews were conducted by master's-level graduate students with experience in clinical interviewing. All interviewers had completed at least two graduate-level courses in psychological assessment. Diagnosticians were trained by observing videotaped interviews and were required to arrive at 100% agreement on at least two observed interviews before conducting an ADIS on their own.

Youth reporting high CSRs for non-PTSD diagnoses, but not for PTSD, were excluded from this study; however, their families were provided with referral information. Further, children taking medications were not excluded from this study. John's guardian reported that he was on medication for encopresis (he began the medication in January 2009) and according to her mother, Elizabeth has been taking a mood stabilizer since age 7. Thus, neither started medication soon before or during the intervention. None of the six participants in this sample was receiving concurrent psychosocial treatments during the *StArT* manual.

Assessment of Posttraumatic Stress Symptoms

PTSD symptoms were measured through the Reaction Index for Children (PTSD-RI; Frederick, Pynoos, & Nadar, 1992). The RI was developed as an interview for the diagnosis of PTSD, has been adapted into a self-report questionnaire (Frederick et al., 1992), and strong test–retest reliability estimates of the RI have been reported in disaster survivors ($r = .93$, $p < .05$; Goenjian et al., 2001). Similar to previous research (Vernberg, LaGreca, Silverman, & Prinstein, 1996) the RI used for this study contains 20 items, with answer choices modified for ease of administration from the original five options to three options: 0 (*none of the time*), 2 (*some of the time*), and 4 (*most of the time*). Total RI scores thus range from 0 to 80. Classification of severity of symptoms were based on the groups developed by Frederick et al. (1992), which includes the categories: doubtful (score of 0–11), mild (12–24), moderate (25–39), severe (40–59), and very severe (60–80). The RI appears sensitive to treatment effects in samples of youth hurricane survivors (Salloum & Overstreet, 2008). Youth were administered the PTSD-RI as part of the pretreatment screening assessment, weekly at the baseline assessments, weekly during treatment, and at the posttreatment assessment.

The parent report assessment also included the Diagnostic Interview Schedule for Children-Predictive Scales (DISC-PS; Lucas et al., 2001). The DISC-PS was used to minimize parent burden and was conducted during the parent phone interview to measure and identify parent reporting of other anxious and depressive symptoms in their children (i.e., simple phobia, social phobia, agoraphobia, panic disorder, GAD, obsessive-compulsive disorder, and major depressive disorder) and other types of psychopathology (i.e., schizophrenia, ADHD, oppositional defiant disorder, conduct disorder, and substance use). Parents responded to DISC-PS items by choosing 0 (*no*) or 1 (*yes*). Findings comparing the DISC-PS to the full Diagnostic Interview Schedule for Children indicate excellent specificity and sensitivity of the predictive scales (Lucas et al., 2001). DISC-PS parent scales evidence good test–rest reliability (parent report of children's anxious symptoms, $\alpha = .86–.89$ and depressive symptoms, $\alpha = .78–.82$; Lucas et al., 2001). The total number of DISC-PS items measuring anxious symptoms was summed separately to derive total and anxiety scores at pre- and posttreatment.

Youth reporting additional diagnoses, and/or who did not report PTSD as their primary diagnoses were not excluded from this study. Given research suggesting that youth trauma survivors typically present with a constellation of emotional problems (Bolton, O'Ryan, Udwin, Boyle, & Yule, 2000; Cortes et al., 2005; Kiser, Heston, Millsap, & Pruitt, 1991; Ruchkin, Schwab-Stone, Koposov, Vermeiren, & Steiner, 2002; Yule et al., 2000), it was decided to include youth with multiple diagnoses. Further, there is a gap between intervention research and its application in practical settings, in part because clinicians do not feel that results from treatment studies generalize to their clients (Barlow & Nock, 2009). We thought that youth with multiple diagnoses might be more exemplary of what would be seen in applied or school settings postdisaster, and we wanted our findings to reflect the treatment process in this type of client. However, referral for additional problems was considered when staffing the child with school counselors, and for youth still meeting criteria for other problems after completion of the *StArT* manual appropriate referral was made.

Additional Treatment Outcome Measures

Changes in factors associated with anxiety disorders such as negative cognitions (i.e., cognitive errors, control beliefs) and anxiety sensitivity were also assessed at pre- and posttreatment. Administration of the Childhood Anxiety Sensitivity Index (CASI; Silverman, Fleisig, Rabian, & Peterson, 1991) occurred at pre- and posttreatment. The CASI is an 18-item measure designed to assess children's fear of different symptoms of anxiety. The CASI evidences good test–retest reliability estimates in predominately Caucasian and African American youth (e.g., *r*s in the = .75 to .8 range; Ginsburg & Drake, 2002; Silverman et al., 1991) and is sensitive to treatment effects in anxiety-disordered youth (Ollendick, 1995). The Children's Negative Cognitive Error Questionnaire (CNCEQ; Leitenberg, Yost, & Carroll-Wilson, 1986) assessed cognitive errors pre- and posttreatment. The CNCEQ is a 24-item questionnaire with items composed of hypothetical vignettes and a negative interpretation. Individuals indicate how closely the interpretation described coincides with their own. Good reliability and internal consistency estimates, $\alpha = .71–.60$ (Leitenberg et al., 1986; Thurber, Crow, Thurber, & Woffington, 1990), as well as sensitivity to treatment effects in anxiety-disordered youth (Silverman, Kurtines, Ginsburg, Weems, Rabian, et al., 1999), have been found.

Children's control beliefs were measured pre- and posttreatment using the short form of the Anxiety Control Questionnaire for Children (ACQ-C; Weems, Silverman, Rapee, & Pina, 2003). Items on the ACQ-C measure perceived lack of control over anxiety-related events. Findings suggest that the ACQ-C is sensitive to treatment effects in anxiety-disordered youth (Muris, Mayer, den Adel, Roos, & van Wamelen, 2009) and the short version evidences a pattern of test–retest correlations similar to that of the full form (1-year test–retest reliability, $r = .58$), and has good internal consistency estimates, $\alpha = .85$ (Taylor, Costa, Cannon, Adams, & Weems, 2006).

Clinically Significant Change

Evidence of clinically significant change has typically been assessed through showing real-life change in functioning (e.g., return to attending school/work regularly, improvement in school grades, no longer behaviorally avoiding feared object) or through reduction in symptoms from non-normative to normative levels at the end of therapy (Kendall & Norton-Ford, 1982; Nietzel & Trull 1988). For this study, clinically significant change was indexed by no longer meeting diagnostic criteria for PTSD on the ADIS-C/P PTSD interview and by reduction of PTSD symptoms on RI into normative range at posttreatment assessment (i.e., an RI score in the doubtful range (0–11).

PROCEDURES

This project received approval from the Institutional Review Board at the University of New Orleans. In order for youth to participate in treatment, parents provided consent for their children's participation in treatment; youth were asked to provide their assent. The lead investigator obtained consent, and administered the PTSD scale of the ADIS-IV and DISC Predictive Scales to parents through phone contact. After consent and assent were obtained, youth pretreatment assessment sessions were scheduled. Pre- and posttreatment assessment was conducted by a master's-level graduate student and included the ADIS-IV and other pretreatment measures (the CASI, CNCEQ, ACQ-C).

Weekly treatment sessions occurred in a designated room and were conducted by a doctoral-level graduate student (the first author) with experience in clinical interviewing, assessment, and therapy with youth.

TREATMENT OVERVIEW

The *StArT* treatment protocol is a manual-based 10-session (each lasting approximately an hour) intervention designed for traumatized youth and consisting of five main components described below (Saltzman et al., 2007; see also Allen et al., 2006). The treatment manual is cognitive–behavioral in orientation, and in the tradition of CBTs, youth are assigned therapeutic homework tasks at the end of each session. A typical session began with the therapist reviewing the main points of the previous session, going over homework assigned during the previous session, and setting the agenda for the current session (e.g., describes the session objectives). After introducing session objectives, the therapist discussed prescribed session material with the child. The therapist facilitated child engagement in sessions through the use of handouts and other therapeutic activities (e.g., writing and art activities) stipulated by the *StArT* manual. The main treatment modules and an abbreviated description of session-by-session material is provided next.

Psychoeducation

During psychoeducation, the therapist built rapport with the child through normalization of symptoms and problems (e.g., lots of kids and adults change after they have gone through scary things). In Session 1, children were provided psychoeducation regarding posttraumatic stress symptoms and responses. Children were given homework designed to track distressing, trauma-related reactions/behaviors most problematic to them for Session 2. Session 2 began with a review of Session 1 homework, the introduction of the cognitive–behavioral conceptualization of anxiety, and the importance of approach behavior. To encourage approach behavior, youth were taught to use the acronym "STIC," or to "show that I can" face my fears. During Session 2, youth were also taught breathing and muscle relaxation skills and how to identify other activities that can be done to reduce anxiety (e.g., read a book, play outside, go for a bike ride). At the end of Session 2, the therapist assigned STIC tasks and use of relaxation techniques for homework.

Cognitive Restructuring

In Session 3, cognitive restructuring strategies were introduced to help youth cope with troubling, trauma-related thoughts. The first strategy is called "STOP" (recognize you feel Scared, you look at the Thoughts that make you feel scared, you try to come up with Other thoughts, you Praise yourself). The child and therapist practiced using STOP in session. Youth were asked to use the STOP technique for homework. In Session 4, the therapist taught the child other types of cognitive restructuring methods for coping with distressing thoughts and feelings. Youth were taught to monitor fearful thoughts and reframe them through the use of "it is possible versus it is likely" (e.g., it is possible that a storm in the Atlantic may turn into a hurricane, but it is not necessarily likely).

Exposure

Exposure began in Session 5. The therapist introduced and helped the child to define and discuss trauma and loss reminders while assigning fear ratings to them. The therapist and child developed a list of helpful coping skills for use when confronted by trauma and/or loss cues. In Session 6, the therapist helped the child begin to construct a trauma narrative. The therapist began by providing the child with a rationale for narrative development (i.e., it helps gain control of memories so that reminders are not so painful) and then helped the child choose an event for narrative construction. Events chosen for narrative development included incidents directly witnessed during the hurricane, and/or hurricane-related memories causing a lot of distress/problems over the course treatment. The role of the therapist was to know when and how to increase and decrease the child's level of engagement during the narrative (i.e., going fast vs. slow through some parts of the narrative, keeping the child grounded, and helping him or her relax when the process gets intense). The therapist helped the child to demarcate the intensity of these events. In Session 7, the child retold the narrative, with the therapist helping the child to identify hurtful or nonhelpful thoughts about things that occurred during the traumatic event. The therapist helped the child challenge and cognitively reframe feelings of loss, shame, or blame through the cognitive restructuring strategies covered in Sessions 3 and 4. In Session 8, the narrative is retold to address the child's misunderstandings of what happened (e.g., child's feelings of self-blame or shame) in terms of other people's actions during the storm. For example, the therapist helped children process anger or sadness toward their parent for refusing to allow them to evacuate with all their toys (e.g., there was not enough room for all the child's toys in the family car).

Problem Solving
In Session 9, the therapist taught the child problem-solving skills through facilitating the identification, development, and prioritization of the current problems list. Then, the therapist introduced the ABC's of problem solving (i.e., Ask if the problem belongs to you, Brainstorm many possible solutions, and Choose the best ones). The therapist helped the child identify the problems that *are not* a kid's responsibility to fix (e.g., parents arguing with relatives, parents not having enough money). Then, children identified one or two problem situations they might encounter before Session 10. For homework, the child was asked to implement the problem-solving model during these situations.

Relapse Prevention
In Session 10, the last session, the therapist taught the child relapse prevention techniques and discussed the child's treatment experience with him or her. The therapist began the session by reviewing the child's treatment experience and asking the child how his or her life changed over the course of treatment. Then the therapist asked the child to anticipate potential challenges that might occur in the next few months (e.g., anniversaries, holidays, hurricane season). The therapist helped the child create a list of challenging events that might occur in the next 6 months and initiated a discussion with the child regarding effective ways of coping with these events. Upon completion of the list, the therapist introduced the concept of "slipping," (e.g., times when the child may feel he or she has fallen back to where he or she was pretreatment) and how "slips" do not mean that the child is back where he or she started. When slips occur, the child should pick him- or herself back up, try not to get frustrated and give up, and continue to make progress. At the close of Session 10, the therapist facilitated the goodbye process by reminding the child of all the positive changes he or she has made, and expressed his or her admiration for the child's courage and hard work.

TREATMENT FIDELITY

A checklist of session-by-session objectives was developed based on the treatment manual. The therapist used these checklists as guidelines for sessions to ensure treatment fidelity. Two parents consented to having their child's treatment sessions audiotaped and assessment of treatment fidelity was examined by comparing session manual content (through fidelity checklists) with information covered in the audiotaped sessions. Two trained observers listened to all treatment sessions. Observer agreement was measured through calculation of percent agreement between raters (i.e., the number of times raters agreed on fidelity checklist items was divided by the total number of checklist items). Across all sessions fidelity to treatment content and session goals were 98% for Sarah and 94% for Elizabeth, suggesting adherence to the manual content.

Intervention Design
This study employed a partially nonconcurrent MBL (see also Lumpkin et al., 2002). Whereas a concurrent MBL calls for beginning the baseline assessment of all participants at a single time point with differing baseline lengths (for each participant or behavior of interest) thereafter, the nonconcurrent alternative allows for the continuous assessment of participants as consents are obtained. After pretreatment assessment, participants were assigned a baseline length to treatment. However, an issue with using a purely nonconcurrent MBL is that it does not control for the effects of history. Thus, when possible, two participants were assigned to start baseline at the same time (i.e., for some sets of participants the baseline periods were concurrent) to control for historical factors and to demonstrate replication of findings across participants. In this sample, participants were randomly assigned to baselines. Jennifer and Kelly were assessed concurrently with Jennifer assigned a 2-week wait and Kelly a 4-week wait. John and Michael were assessed concurrently and assigned a 3-week wait. Sarah and Elizabeth were assessed nonconcurrently with Sarah assigned a 3-week wait period and Elizabeth assigned a 4-week wait.

Results

Weekly summed RI scores were graphed and visually inspected for trends as the main method of evaluating treatment efficacy (see Figure 1) given the MLB design. Overall, the pattern of results suggests declines in PTSD symptoms over time, which is consistent with the expectation of reductions in symptoms following initiation of the intervention.[1] However, there was substantial

[1] To evaluate the impact of treatment components on specific types of PTSD symptom cluster symptoms (avoidance and numbing, hyperarousal, and reexperiencing symptoms), graphic representation of individual reduction patterns for PTSD symptom clusters (taking the mean of PTSD cluster items for comeasurability) were visually inspected. Visual inspection of data indicate downward trends in each RI symptom cluster score across treatment (T1 to T10) and at posttreatment assessment (T11) similar to total scores and suggesting little or no unique variation in subsymptoms (avoidance and numbing, hyperarousal, and reexperiencing).

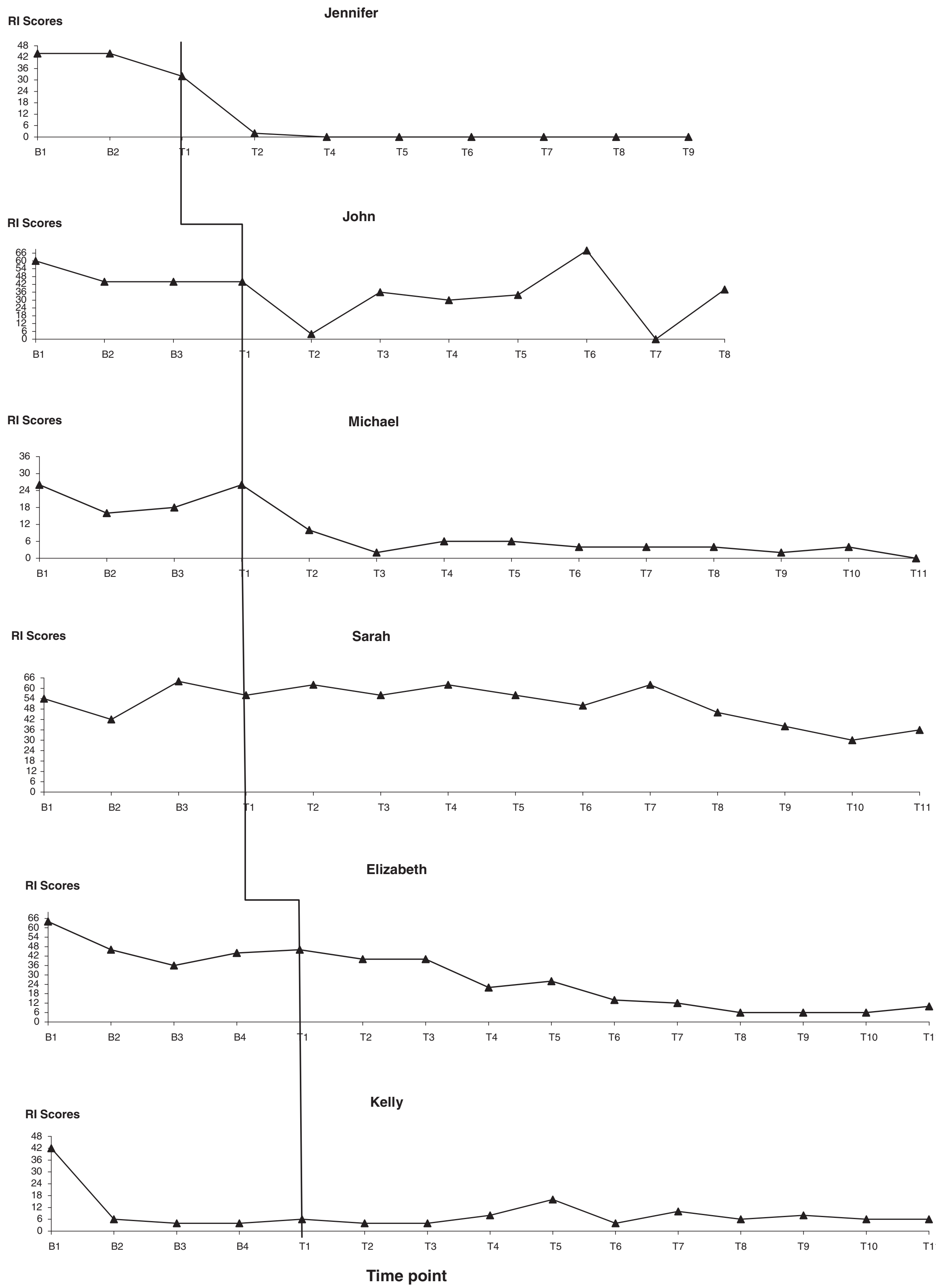

FIGURE 1 Individual PTSD symptom change across treatment sessions (summed total scores).

individual variation in both treatment response and baseline stability. The pattern of results from Jennifer, Michael, and Elizabeth were highly consistent with expectations that the initiation of therapy would be associated with decreases preceded by a stable baseline. Sarah's declines occurred later in the intervention, Kelly's occurred at the second baseline assessment, and a consistent pattern of treatment response is difficult to discern from John's data but a conservative view would suggest that he was either inconsistent in his treatment ratings or did not benefit from the intervention. This interpretation is consistent with his termination of therapy at Session 7 (T8 is John's posttreatment assessment data). A stable baseline is characterized by relatively little variability and the absence of slope (Kazdin, 2003); however, assessments using self-report of symptoms often show more variation than behavioral observations (Lumpkin et al., 2002).

Visual inspection of data indicates unique variation in baseline PTSD symptoms. However, Jennifer, John, Michael, Elizabeth, and Sarah reported RI scores ranging from moderate to very severe across baseline assessments indicative of general stability in baseline symptoms. In sum, visual inspection of the data in Figure 1 suggest that for Jennifer, Michael, Sarah, and Elizabeth the downward trends in RI scores across treatment (T1 to T10) and at posttreatment assessment (T11) indicate possible treatment efficacy. John showed stability in RI scores until intervention and symptom fluctuation with no identifiable trend during intervention. Visual inspection of John's intervention data did not indicate treatment efficacy at the individual level. Kelly reported scores ranging from the severe to doubtful across baselines and thus her baseline data were not stable enough to make the prediction that, without intervention, she would report RI scores in the moderate range.

A reliable change index (RCI; Jacobson & Traux, 1991) was also calculated to detect how much change had occurred in RI scores (i.e., PTSD symptoms) during the course of treatment. Five youth in this sample reported reliable change in symptoms; however, John's symptoms did not show reliable change.

CLINICALLY SIGNIFICANT CHANGE

Clinically significant change was defined as no longer reporting diagnosis of PTSD at posttreatment assessment and reduction in symptoms on the RI to the doubtful range. Table 1 displays pretreatment and posttreatment diagnoses and symptom levels. As shown in Table 1, all six participants no longer met criteria for PTSD at posttreatment assessment. Fifty percent of youth (Jennifer, John, and Kelly) did not meet criteria for any diagnoses upon concluding treatment. Michael, Sarah, and Elizabeth reported diagnoses at posttreatment assessment. Elizabeth reported reductions in severity of diagnosis for social phobia (at pretreatment, 7; at posttreatment, 4). Sarah also indicated reductions in severity of diagnoses for dysthymia (at pretreatment, 7; at posttreatment, 5) and for GAD (at pretreatment, 7; at posttreatment, 4). Report of diagnoses for Michael and Sarah were not consistent across ADIS-C administrations in that Michael reported diagnoses of social phobia and Sarah reported specific phobia at posttreatment assessment. Table 2 also shows score categories (doubtful, 0–11; mild, 12–24; moderate, 25–39; severe, 40–59; and very severe, 60–80) for individual RI scores reported at pre- and posttreatment assessment. Four children went from reporting RI scores in the moderate range (Michael) and severe ranges (Jennifer, Elizabeth, Kelly) at pretreatment to reporting scores in the doubtful range at posttreatment. While not all youth reported scores in the doubtful range after treatment, all youth reported reductions in PTSD symptoms. John reported RI scores in the severe range at pretreatment and in the moderate range at posttreatment (a 6-point reduction). Sarah reported RI scores in the severe range at pretreatment and in the moderate range at posttreatment (an 18-point reduction).

GROUP LEVEL REDUCTIONS IN SYMPTOMS

To further investigate improvement, Wilcoxon signed rank tests, and calculation of Cohen's d were conducted on the treatment outcome measures and results are shown in Table 2. As shown in Table 2, the RI scores and CNCEQ scores significantly decreased from pre- to posttreatment ($M = 15$, $SD = 17.5$), $t(5) = 4.4$, $p = .007$. There were no statistically significant differences in CASI, ACQ-C, and DISC-PS total and anxiety scores from pre- to posttreatment. However, examination of the means and standard deviations for the CASI, ACQ-C, and DISC-PS total and anxiety scores suggest changes in the expected direction for each participant's scores except one on each of these measures (i.e., John's CASI, and DISC-PS anxiety and total scores increased; Jennifer's ACQ-C score decreased).

Discussion

The results from this study suggest that the *StArT* manual (Saltzman et al., 2007), a trauma focused CBT for youth disaster survivors, may be helpful in the reduction of child-reported posttraumatic stress symptoms. The results add to the growing support for CBT interventions in youth following exposure to disaster (e.g., Chemtob et al., 2002; Salloum & Overstreet, 2008). Whereas the *StArT* components are empirically based, this study represents the first systematic evaluation. The pattern of results from Jennifer, Michael, and Elizabeth were highly

Table 2
Reductions in PTSD-RI and Treatment Outcome Scores and Treatment Effects

	Pretreatment Scores	Posttreatment Scores	Wilcoxon *z*	*d*
PTSD-RI[a] (*M*, *SD*)	45.0(11.6)	15.0(17.5)	2.20**	2.00
Jennifer	44[d]	0[b]		
John	44[d]	38[c]		
Michael*	26[c]	0[b]		
Sarah	54[d]	36[c]		
Elizabeth	60[e]	10[b]		
Kelly	42[d]	6[b]		
CASI[f] (M, SD)	30.8(8.10)	26.0 (7.22)	1.36	.625
Jennifer	22	21		
John	26	36		
Michael	24	20		
Sarah	38	33		
Elizabeth	42	27		
Kelly	33	19		
ACQ-C[g] (M, SD)	14.5(10.2)	23.0(10.6)	–1.59	–.817
Jennifer	30	23		
John	13	20		
Michael	21	28		
Sarah	1	19		
Elizabeth	7	8		
Kelly	15	40		
CNCEQ[h] (M, SD)	47.0(25.1)	31.7(7.39)	2.20**	.826
Jennifer	44	36		
John	41	32		
Michael	37	25		
Sarah	97	44		
Elizabeth	35	28		
Kelly	28	25		
DISC[i] (M, SD)	26.8(21.6)	21.6(19.2)	1.48	.254
Jennifer	13			
John	9	12		
Michael	30	18		
Sarah	33	23		
Elizabeth	58	53		
Kelly	4	2		
ANX[j] (M, SD)	11.2(9.54)	7.40(7.70)	1.63	.438
Jennifer	5			
John	1	2		
Michael	13	5		
Elizabeth	23	20		
Sarah	17	9		
Kelly	2	1		

Note. [a]Reaction Index total scores; RI categories: [b]Doubtful; [c]Moderate; [d]Severe; [e]Very severe; [f]Childhood Anxiety Sensitivity Index; [g]Anxiety Control Questionnaire for Children; [h]Children's Negative Cognitive Error Questionnaire; [i]DISC-PS total scores; [j]DISC-PS total anxiety scores; *reported an RI score of 40 at screening; ** $p < .05$.

consistent with the expectation that the initiation of therapy would be associated with decreases in PTSD symptoms preceded by a stable baseline. Moreover, at posttreatment assessment, none of the youth from this sample reported a diagnosis of PTSD, and half of youth reported no anxiety disorder diagnoses. It is important to note that for four of the six youth PTSD was not the most severe anxiety problem. This is not surprising since multiple forms of emotional disruption is common in disaster-exposed youth (La Greca et al., 2002). At posttreatment assessment, there was a reduction in the incidence of these disorders and the severity of most of these diagnoses decreased from pre- to posttreatment. Further, whereas there were no statistically significant differences in CASI, ACQ-C, and DISC-PS total and anxiety scores from pre- to posttreatment, examination of individual scores suggest changes in the expected direction for each participant except one on each of these measures (i.e., John's CASI and DISC-PS total and anxiety scores increased; Jennifer's ACQ-C score decreased). Scores on the CNCEQ did show statistically significantly reductions but given the small sample additional research is necessary to establish the validity and clinical implications of findings regarding the effects of the *StArT* manual on cognitive errors.

Though results indicate a reduction of symptoms for most participants, a conclusion that the intervention was solely or invariably responsible for symptom decline does not seem justified as one participant reported significant reductions at baseline (Kelly) and the intervention failed to produce systematic decrease in one other case (John). It is also possible that trauma history influenced participant response to intervention. The *StArT* manual was designed to help youth exposed to hurricanes and participants who had more complex trauma histories (John, history of neglect; Sarah parental loss) did show less than expected PTSD symptom reduction (Figure 1).

This study responds to the general call for further study of treatment change using idiographic designs (Barlow & Nock, 2009; Kazdin, 2008); findings were consistent with literature using multiple baseline designs to evaluate treatment efficacy for youth with anxiety disorders (e.g., Lumpkin et al., 2002; Ollendick, 1995) and traumatized youth more specifically (e.g., Feather & Ronan, 2006; Saigh, 1987). However, future examination of the *StArT* manual might benefit from the inclusion of separate multiple baseline evaluations of treatment components through the assessment of the skills purportedly learned during each of the modules. For example, multiple baseline evaluation has been used to illustrate session-level changes relative to treatment components by showing that skills (e.g., cognitive coping skills) learned upon component introduction increased while PTSD symptoms decreased (Feather & Ronan, 2006).

Although the goal of this study was to provide idiographic evaluation, conducting replications of the present study, or randomized control trials of the *StArT* manual in large samples of youth, appear justified and important. The results of this study were generally positive; however, data on the generalizability of this treatment is still needed. It is notable that comparison of effect sizes generated in this study with effect sizes based on change in PTSD due to time alone from previous longitudinal studies (e.g., youth survivors of the Spitak earthquake, $N = 89$ surveyed at 18-month follow-up, estimated normal change as a function of Cohen's $d = .55$; Goenjian et al., 2005; youth survivors of Hurricane Andrew, $N = 442$ surveyed at 3- and 10-month follow-up, estimated Cohen's $d = .58$, La Greca et al., 1996) suggest that the average decrease was larger than would be expected by time alone and reductions were comparable or larger than previous treatment studies of youth exposed to Katrina (Cohen's $d = 1.16$ for the PTSD-RI in Salloum & Overstreet, 2008). Conducting growth curve modeling might also indicate the typical symptom change trajectories of hurricane-exposed youth, and further inform efficient intervention strategies for youth surviving natural disasters.

The results also highlight single-case designs as viable and potentially practical methods for school-based intervention evaluation. In terms of providing feasibility information for future work, findings from this study suggest that the *StArT* manual can be administered in school settings. Research suggests that minority youth may experience obstacles to receiving anxiety disorder treatment (Chavira, Stein, Bailey, & Stein, 2003) and school settings may provide a useful context for minority youth to receive intervention. However, one logistic issue is space to conduct treatment sessions (Mufson et al., 2004; Pincus & Friedman, 2004). For schools participating in this study, finding a private space to have sessions was at times difficult due to limited space for adjunct services. In order to expand mental health services within the school setting, future work might benefit from evaluation of the *StArT* manual in group format rather than the individual format used here. Reserving one location within the school for a weekly group may be easier to achieve than coordinating several individual sessions during the week. Research indicates that traumatized youth who receive school-based treatments in either group or individual format show substantial and significant reductions in PTSD symptoms (Chemtob et al., 2002; Salloum & Overstreet, 2008). Moreover, future implementation of the *StArT* manual within schools might follow a multitiered approach to treatment delivery (e.g., Cohen et al., 2009; Layne et al., 2008). Expanding the framework of the intervention to include school and community mental health professionals, parents, and teachers may facilitate the generalization and maintenance of treatment components.

The results from this study provide promising outcomes, but are not without limitation. First, one child from this sample (Kelly) did not maintain symptom stability at baseline. When baseline data shows variability, it is difficult to draw conclusions about intervention effects. For cases in which baseline data is not stable, the causal role of the intervention cannot be assumed given symptom reduction might be occurring for other reasons (e.g., history or maturation; Kazdin, 2003). Thus, downward trends in Kelly's baseline data could suggest that symptom reduction would have occurred over time, without intervention. Second, neither extensive life history of problems nor inter-rater reliability were assessed across ADIS administrations. Sarah and Michael reported new diagnoses at posttreatment (for Sarah, specific phobia, and for Michael, social phobia; see Table 2). Some of the response variability across ADIS-IV administrations may be related to emotion-related deficits found in anxiety disordered youth. Researchers have theorized that anxiety disordered youth have difficulties understanding and managing their own emotional experiences (Suveg, Kendall, Comer, & Robin, 2006). It is plausible that participating in treatment improved Sarah's and Michael's understandings of the types of situations and experiences that are anxiety provoking to them, and this perhaps accounts for the variance in reported diagnoses. Further, only one therapist conducted the intervention in this sample and thus research is needed to assess generalizability of the intervention across multiple therapists.

Finally, generalizability of our findings is also limited by sample composition and size. However, intervention generalizability was not the goal of this initial evaluation. Future investigations of the *StArT* manual might include a larger and more ethnically and socioeconomically diverse group of youth. Findings are also limited by the lack of follow-up data. While initial single-case design evaluations of interventions often lack follow-up (Nakamura, Schiffman, Lam, Becker, & Chorpita, 2010; Ross & Horner, 2009) no conclusions about the maintenance of treatment can be drawn from this study and so future research is needed to examine whether treatment gains can be maintained. Assessment of treatment satisfaction, and intervention impact on related outcomes such as social and academic problems would strengthen the study findings and should be assessed in more comprehensive evaluations of the *StArT* manual.

The apparent chronic nature of mental health problems following Katrina (Kessler et al., 2008; Weems et al., 2010) speak to the importance of long-term community and also school-based efforts to address the mental health needs of individuals following natural disaster that are ongoing and not just short term immediately postdisaster (Drury et al., 2008; Weems & Overstreet, 2008). The *StArT* manual evaluated here has promising potential for underserved youth. Funding large-scale implementation of this or similar programs in areas hard hit by a disaster may help reduce the mental health burden on youth reducing rates of PTSD, fostering symptom decline, and potentially targeting traditionally underserved communities.

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